# Unleashing the Intrinsic Visual Representation Capability of Multimodal Large Language Models

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## Abstract

*Multimodal Large Language Models (MLLMs) have demonstrated remarkable proficiency in multimodal tasks. Despite their impressive performance, MLLMs suffer from the modality imbalance issue, where visual information is often underutilized compared to textual representations in deeper layers, leading to degraded visual performance or hallucinations. This issue stems from the predominant reliance on next-text-token-prediction during training, which fails to provide direct visual supervisory signals, resulting in progressive homogenization of visual representations throughout the layers. To this end, we propose **La**tent Visual **Re**const**r**uction (**LaVer**), a novel training framework that facilitates MLLMs in learning more discriminative visual representations via masked image modeling in the joint latent semantic space of LLM. Our method offers direct visual activation to MLLMs, which exhibit increased visual attention allocation, indicating enhanced utilization of visual information. Extensive experiments across diverse benchmarks prove the superiority of our approach in various scenarios, especially those requiring dense visual capabilities.The code of LaVer is available at https://github.com/Fir-lat/LaVer.*

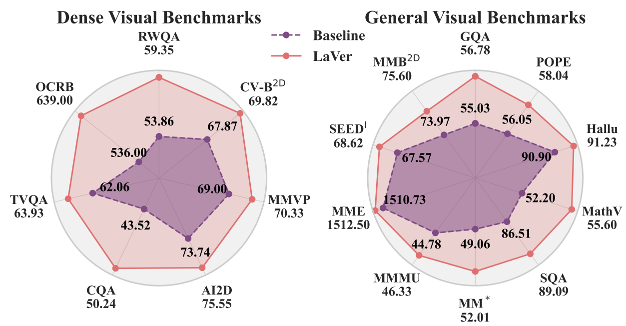


Figure 1. **Benchmark Performance.** LaVer consistently outperforms the baseline across diverse benchmarks, especially on dense visual tasks such as OCRB [66] and CQA [73]. The results are obtained with SigLIP 2 [100] and Qwen2.5-7B-Instruct [86].

## 1. Introduction

Multimodal Large Language Models (MLLMs) [1, 2, 6, 7, 24, 55, 61–63, 96] have emerged as a transformative paradigm for real-world multimodal tasks, fusing the linguistic reasoning power of Large Language Models (LLMs) [5, 11, 41, 81, 87, 88, 99, 112, 113] with the advanced cognition and perception abilities of vision models [82, 89, 92, 100, 119]. MLLMs have demonstrated significant potential in enhancing visual capabilities across a wide spectrum of tasks, ranging from visual question answering to complex multimodal reasoning [3, 37, 56, 123].

Despite their impressive capabilities, MLLMs are afflicted by the *modality imbalance* issue [14, 122, 126], where the model exhibits a systematic bias towards textual information in multimodal tasks. MLLMs frequently generate confident responses even when presented with absent or incongruent visual inputs, as empirically demonstrated in [68, 83, 122, 126]. When rich visual information is available, MLLMs still tend to produce outputs predominantly grounded in textual context [47, 52]. Qualitative analysis reveals that MLLMs allocate substantially more attention to textual content than vision tokens [110], as the text modality demonstrably contributing more to the predictions [83].

The consequences of this modality imbalance are multifaceted. Model outputs become systematically biased toward the dominant textual modality, fundamentally failing to leverage the full potential of underrepresented visual information [90, 103], leading to degraded performance across various benchmarks [64, 85, 126] and increased visual hallucinations [42, 52, 53, 91, 107], ultimately diminishing the reliability [68, 97] of MLLMs. Addressing this challenge is crucial, as ideal MLLMs should seamlessly integrate and leverage the full potential of each modality to achieve robust multimodal understanding [20, 97].

The modality imbalance stems from the fundamen-

*Work done during internship at Tencent Hunyuan Group.